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St. Joseph's college f Arts & Science (Autonomous)

St. Joseph's College Road, Cuddalore-607001

B. Sc DEGREE EXAMINATION-APRIL 2021

BC101-BIOMOLECULES-I

Time: 3 hrs **Max Marks: 75**

SECTION-A (25X1=25)

Answer ALL the questions:

Choose the correct answer: (10X1=10)

1. Complete transfer of one or more electrons from one atom to different atom forms

   a)ionic bonds b) covalent bonds

   c)metallic bonding d) co-ordinate bonding

2. Which of the following groups are all polysaccharides?

   a) sucrose, glucose and fructose

   b) cellulose and starch and fructose

   c) glycogen, sucrose and maltose

   d) glycogen, cellulose and starch

3. Glucose is a ---------- sugar

   a) triose b) tetrose

   c) pentose d) hexose

4. Sucrose contains----------

   a) glucose and fructose b) glucose and galactose

   c) glucose and glucose d) galactose and fructose

5. DNA rich in G-C pairs have

   a) one hydrogen bond b) two hydrogen bonds

   c) three hydrogen bonds d) four hydrogen bonds

6. Osazones are not formed with the
   a) glucose b) fructose
   c) sucrose d) lactose
7. Nucleotide composed by
   a) Base-sugar phosphate b) Base-sugar OH
   c) Sugar phosphate d) Base-sugar
8. Double helical structure of DNA was determined by
   a) Griffith b) Watson and Crick
   c) Hershey and Chase d) Mendel
9. What is the main function of tRNA in relation to protein synthesis
   a) Inhibits protein synthesis b) Proof reading
   c) transports aminoacids to ribosomes d) All of these
10. Bile pigments
   a) removed with feces b) permanently stored in gall bladder
   c) contains enzymes d)absorbed by large intestine

II. Say True or False (5X1=5)

11. Sugar phosphates are the type of lipid moved through the blood
12. The bile salts acts to emulsify fat
13. A compound with low pH is called an acid.
14. Dehydration joins monomers into polymers for carbohydrates synthesis
15. All glycoproteins are synthesized in ribosomes

III. Answer in one sentence (10X1=10)

16. Define polar and non- polar bonds.
17. What do you mean by polysaccharides?

18. Explain ionic bond.
19. Define chitin.
20. What is the role of insulin in biological system?
21. Define anomers.
22. Define rRNA.
23. What is nucleic acid?
24. Define heme?
25. What is indole?

SECTION-B (4X5=20)

Answer any FOUR questions:

26. Discuss about covalent and non-covalent bonds with examples.
27. Write about epimers and mutarotation.
28. Discuss structure and biological importance of polysaccharides.
29. Explain nucleotides and nucleosides.
30. Discuss biological importance of pyridine and pyrrole.
31. Describe Kiliani synthesis.

SECTION-C (3X10=30)

Answer any THREE questions:

32. Discuss the nature and types of chemical bonding
33. What are polysaccharides? Explain their types with examples.
34. Explain double helical structure of DNA in detail.
35. Describe the biological importance of structural polysaccharides
36. What is heme? Explain its synthesis.

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